RaceDay Event Management System

API Endpoint Plan

Module: PROG6212 – Programming 2B

Assessment: Portfolio of Evidence, Part 1

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Role Summary:

Role	Responsibilities in the API
Organiser	Creates and maintains events, defines the distance categories offered by each event, views the full entrant list for events they own, and captures finish times and finishing positions after a race.
Participant	Registers an account, browses published events, enters a chosen category, views and withdraws from their own enrolments, and views their own results.
None	Indicates a public endpoint that requires no authentication; used for registration, login and browsing the published event list.
Any	Indicates the caller must be signed in, but either role is acceptable.

Authentication:

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/auth/register	Register a new user as Organiser or Participant	None	{firstName, lastName, email, password, phoneNumber, dateOfBirth, role}	201 Created; 400 invalid data; 409 email exists
POST	/api/auth/login	Authenticate and start a session	None	{email, password}	200 OK + user/role; 401 invalid credentials
POST	/api/auth/logout	End the current session	Any	None	200 OK; 401 not authenticated

User Profile:

Methods	Route	Description	Role	Request Body	Expected Response
GET	/api/users/me	View own profile	Any	none	200 OK; 401
PUT	/api/users/me	Update own profile details	Any	{firstName, lastName, phoneNumber, dateOfBirth}	200 OK; 400; 401

Events:

Method	Route	Description	Role	Request Body	Expected Response
GET	/api/events	List all published events	None	none	200 OK
GET	/api/events/{id}	View one event with its distance, event type and its categories	None	none	200 OK; 404 not found
POST	/api/events	Create a new event	Organiser	{eventName, description, eventDate, distanceKm, eventType, venueId}	201 Created; 400; 401;403
PUT	/api/events/{id}	Update an event the organiser owns	Organiser	{eventName, description, eventDate, distanceKm, eventType, venueId, status}	200 OK; 400; 403; 404
DELETE	/api/events/{id}	Delete an event	Organiser	none	204 No Content; 403; 404; 409 has enrolments

Note on the Event model: each event records its own distanceKm and eventType. The eventType value is restricted to Run, Walk or Cycle and is validated by the API before the

record is saved. The distanceKm on the event is the headline distance advertised for the race, while the distanceKm stored on each category is the specific distance a participant actually enters, for example a 5km or 21.1km option within the same event. The two values serve different purposes and both are kept.

Categories:

Method	Route	Description	Role	Request Body	Expected Response
GET	/api/events/{eventId}/categories	List categories for an event	None	none	200 OK; 404
POST	/api/events/{eventId}/categories	Add a category to an event	Organiser	{categoryName, distanceKm, entryFee, maxParticipants}	201 Created; 400; 403; 409 duplicate name
PUT	/api/categories/{id}	Update a category	Organiser	{categoryName, distanceKm, entryFee, maxParticipants}	200 OK; 400; 403; 404
DELETE	/api/categories/{id}	Remove a category	Organiser	none	204; 403; 404; 409

Enrolments:

Methods	Route	Description	Role	Request Body	Expected Response
POST	/api/enrolments	Enter a chosen event category	Participant	{categoryId}	201 Created; 400; 401; 403; 409 already enrolled or full
GET	/api/enrolments/me	View own enrolments	Participant	none	200 OK; 401

PUT	/api/enrolments/{id}	Change selected category before closing date	Participant	{categoryId}	200 OK; 403; 404; 409
DELETE	/api/enrolments/{id}	Withdraw from an event	Participant	none	204; 403; 404
GET	/api/events/{eventId}/ enrolments	View all entrants for an event	Organiser	none	200 OK; 403; 404

Results:

Methods	Route	Description	Role	Request Body	Expected Response
POST	/api/results	Capture finish time and position	Organiser	{enrolmentId, finishTime, position}	201 Created; 400; 403; 409 result exists
PUT	/api/results/{id}	Correct a captured result	Organiser	{finishTime, position}	200 OK; 400; 403; 404
GET	/api/results/me	View own results	Participant	none	200 OK; 401
GET	/api/events/{eventId} /results	View full results list for an event	Any	none	200 OK; 404